

Chapter 30: Cesarean Delivery and Peripartum Hysterectomy — Revision Notes

Williams Obstetrics 26e, pp. 1405–1474 of the PDF. High-yield numbers are in **bold**. The single table is transcribed from the PDF image (it is missing from the extracted chapter text). 16 of the 22 operative figures are included from the PDF.

Big picture: Cesarean delivery = laparotomy + hysterotomy. It trades higher maternal surgical risk (now and in every later pregnancy) for less perineal/pelvic floor injury and less neonatal trauma. Most technique choices (bladder flap, one vs two layers, peritoneal closure, exteriorization) matter less than prophylaxis: **antibiotics before incision, VTE prophylaxis, oxytocin after birth**.

1. Background and Indications

- US cesarean rate: **4.5% (1970) → 32.9% (2009)**; plateau at **31.9% in 2018**.
- **>85%** of cesareans are for **four reasons**: prior cesarean, **labor dystocia/arrest**, **fetal jeopardy**, **abnormal presentation**. The last three are the main **primary** cesarean indications.
- **Why rates stay high:**
 1. Fewer children → more births are to nulliparas (higher risk)
 2. Rising maternal age (especially older nulliparas)
 3. Widespread **electronic fetal monitoring** (vs intermittent auscultation)
 4. Most **breech** fetuses now delivered by cesarean
 5. Fewer **operative vaginal deliveries**
 6. **Obesity** epidemic
 7. More cesareans (fewer inductions) for **preeclampsia**
 8. **VBAC rate fell from 28% (1996) to 13.3% (2018)**
 9. More **assisted reproductive technology**

Table 30-1. Some Indications for Cesarean Delivery

Maternal	Maternal-Fetal	Fetal
Prior cesarean delivery	Cephalopelvic disproportion	Nonreassuring fetal status
Abnormal placentation	Failed operative vaginal delivery	Malpresentation
Maternal request	Placenta previa or vasa previa	Macrosomia
Prior classical hysterotomy	Placental abruption	Congenital anomaly
Unknown uterine scar type		Abnormal umbilical cord Doppler study
Prior uterine incision extension		Thrombocytopenia
Uterine incision dehiscence		Prior neonatal birth trauma
Prior full-thickness myomectomy		
Genital tract obstructive mass		
Invasive cervical cancer		
Prior trachelectomy		
Permanent cerclage		
Prior pelvic reconstructive surgery		
Prior significant perineal trauma		
Pelvic deformity		
HSV or HIV infection		
Cardiac or pulmonary disease		
Cerebral aneurysm or arteriovenous malformation		
Pathology requiring concurrent intraabdominal surgery		
Perimortem cesarean delivery		

HIV = human immunodeficiency virus; HSV = herpes simplex virus.

2. Definitions

Term	Meaning
Cesarean delivery	Birth of a fetus by laparotomy then hysterotomy . Not used for removal of the fetus from the abdomen after uterine rupture or with abdominal pregnancy .
Postmortem / perimortem cesarean	Hysterotomy in a woman who has just died or whose death is expected soon (Ch 50)
Cesarean hysterectomy	Hysterectomy at the time of cesarean
Postpartum hysterectomy	Hysterectomy shortly after vaginal delivery
Peripartum hysterectomy	Umbrella term for both
Total vs supracervical	Total removes body + cervix (usual); supracervical removes body only. Adnexa usually not removed.
Simple (type I) vs radical	Simple is usual. Radical (uterus + parametrium + proximal vagina) for invasive cervical cancer, or percreta extending to the pelvic sidewall

3. Cesarean Delivery Risks

Overall balance: higher maternal surgical risk (now and in later pregnancies) vs less perineal injury and fewer short-term pelvic floor disorders. For the neonate: **less birth trauma and stillbirth**, but **more initial respiratory difficulty**.

Maternal mortality and morbidity

- **Mortality 2.2 vs 0.2 per 100,000** (cesarean vs vaginal; Clark 2008).
- Meta-analysis: **13 per 100,000** with elective repeat cesarean vs **4 per 100,000** with trial of labor.
- Part of the risk is from the **underlying indication** (e.g., long first stage → fever and PPH too). In China, nonindicated cesarean did not raise mortality over spontaneous vaginal birth.
- **Morbidity ~2× vaginal birth**. Main ones: **infection, hemorrhage, VTE**; also anesthetic complications and adjacent organ injury.
- **Repeat cesareans accrue risk**: abnormal placentation, wound/uterine infection, organ injury, cesarean hysterectomy, transfusion (Ch 31).
- **Pelvic floor**:
 - Cesarean → **less urinary incontinence and pelvic organ prolapse**.
 - **Anal incontinence is not influenced by route** (but operative vaginal delivery raises it → sphincter injury).
 - The pelvic floor advantage **fades as women age**; duration of stress UI protection unclear (NIH 2006).

Neonatal morbidity

- **Fetal injury in ~1% of cesareans. Skin laceration most common**; others: cephalohematoma, clavicular fracture, brachial plexopathy, skull fracture, facial nerve palsy.
- **Highest injury rate: cesarean after failed operative vaginal delivery. Lowest (0.5%): elective cesarean.**
- Parkland: ~1/3 entered spontaneous term labor, and **96%** of them delivered vaginally without adverse neonatal outcome.

Cesarean delivery on maternal request (CDMR)

- US estimate **~5%** (older data).
- Reasons: pelvic floor protection, convenience, **fear of childbirth**, less fetal injury.
- Chinese cohort (>66,000): similar serious maternal morbidity and neonatal mortality. Birth trauma, infection and HIE lower with CDMR, but **RDS higher**.
- NIH conference: insufficient data for most outcomes; more research needed.
- **ACOG: not before 39 weeks**; ideally **avoid in women who want several children** (accruing morbidity).

4. Patient Preparation

Delivery availability

- **No national standard** for decision-to-incision time. The old **30-minute rule**: failure to meet it was **not associated with worse neonatal outcome** (Bloom 2006; systematic review).
- Still, with **acute catastrophic fetal deterioration**, deliver as fast as possible; purposeful delay is inappropriate.
- AAP/ACOG: facilities must be able to start cesarean in a time frame that balances maternal and fetal risk.

Informed consent

- A **process**, not a document: diagnosis, alternatives, goals, limitations, risks. Offer **TOLAC** to suitable candidates. Sterilization/IUD consent can be done at the same time.
- A woman **may decline**; document the reasons and that consequences were explained.

- **Jehovah's Witnesses** (start discussions early in pregnancy; use a checklist of accepted products):
 - Usually decline **red cells, white cells, platelets and plasma** (primary components). Some clotting factors or fractions may be acceptable.
 - Maximize Hb: **iron, folate, ± erythropoietin**. Limit phlebotomy; pediatric tubes.
 - Intraop: prompt atony treatment, topical hemostatics, **tranexamic acid, desmopressin, cell salvage**, normovolemic hemodilution, controlled hypotensive/hypothermic anesthesia, uterine artery embolization, occlusive balloons.

Timing of scheduled cesarean

- Elective delivery **before 39 completed weeks** → appreciable neonatal immaturity morbidity.
- **Criteria supporting fetal maturity:**
 - Ultrasound **before 20 weeks** corroborating GA
 - Fetal heart tones heard for **30 weeks**
 - **36 weeks** since a positive pregnancy test

Preoperative care (ERAS)

- **Solids stopped ≥6 h** before; **clear liquids up to 2 h** before. **No bowel prep.**
- Review recent **hematocrit and indirect Coombs** (if positive, ensure compatible blood).
- **Regional analgesia preferred.**
- **Aspiration prophylaxis:** oral antacid (e.g., **Bicitra 30 mL**) shortly before anesthesia ± **IV H₂-receptor antagonist**.
- **Left lateral tilt** (wedge under the right hip) to prevent aortocaval compression/hypotension.
- Fetal heart: Parkland does a **5-min tracing** before elective cases; at minimum document FHR in the OR.
- **Hair removal** does not lower SSI. If needed, **clip on the day of surgery** (fewer SSIs than shaving).
- **Indwelling bladder catheter** (Parkland): moves the bladder away from the hysterotomy, prevents retention after regional block, measures urine. Can be omitted in stable women (less discomfort, UTI rate unchanged).
- **VTE:**
 - Cesarean VTE rate **8 per 10,000** (~2× vaginal birth).
 - **ACOG: pneumatic compression stockings before cesarean** for all women not already on prophylaxis (stop once ambulating).
 - ACCP: early ambulation only if no risk factors. **RCOG most conservative** (pharmacological prophylaxis for most).

Infection prevention

Antibiotic prophylaxis (cesarean = **clean-contaminated**)

Situation	Regimen
Standard	Cefazolin 1 g IV , single dose
Weight ≥80 kg	2 g (CDC); ACOG accepts 2 or 3 g
Weight ≥120 kg	3 g (CDC)
Redose	Blood loss >1500 mL or surgery >3 h
In labor or ruptured membranes	Add azithromycin 500 mg IV (↓ SSI and endometritis; Tita 2016)
History of MRSA	Add vancomycin 15 mg/kg once
Significant β-lactam allergy (anaphylaxis, angioedema, respiratory distress, urticaria)	Clindamycin 900 mg IV + aminoglycoside 5 mg/kg (900 mg clindamycin also in obese women)

- **Timing: within 60 min before incision** (better than after cord clamping, with no neonatal harm). Emergency: as soon as feasible.

- Regimens without a β -lactam may have higher SSI rates → specialist **penicillin-allergy evaluation** can remove the label.
- **Skin prep**: chlorhexidine-alcohol or povidone-iodine; **data favor chlorhexidine**. Night-before chlorhexidine wipe = no benefit.
- Adhesive incisional drapes may slightly ↑ SSI.
- **Vaginal cleansing** with 1% povidone-iodine (or 4% chlorhexidine if allergic) ↓ endometritis/SSI in some studies (not Parkland practice).
- Possible: **48 h oral antibiotics** after cesarean in **obese** women.
- **Endocarditis prophylaxis** only for cyanotic heart disease (repaired or unrepaired), **prosthetic valves**, prior endocarditis, or cardiac transplant with valve regurgitation. Give 30–60 min before delivery; standard cesarean prophylaxis also covers endocarditis.

Other prevention

- Glycemic control; stop smoking; **intraoperative normothermia**.
- **Vaginal seeding** (swabbing the newborn's mouth with vaginal gauze) is **not encouraged by ACOG** (few data, risk of transmitting infection).

Surgical safety

- Joint Commission protocol: verify documents + **"time out"** (right patient, site, procedure; team introductions, antibiotics, expected duration, anticipated complications).
- **Surgical fire** prevention: assess risk, **let alcohol-based prep dry**, keep ignition sources off the patient.
- **Instrument, sponge and needle counts** before and after; if unreconciled → **portable radiograph**.

5. Cesarean Technique: Laparotomy

Choice of incision

	Pfannenstiel (transverse)	Midline vertical
Cosmesis, incisional hernia	Better cosmesis, fewer hernias (follows Langer lines)	More hernias
Operating space / upper abdomen	Limited	Large space, upper abdominal access
Speed of emergent entry	Slower	Faster
Infection risk	Pus can collect between oblique aponeurosis layers	Favored by some when infection risk is high
Neurovascular structures	Ilioinguinal, iliohypogastric nerves; superficial and inferior epigastric vessels → more bleeding, hematoma, nerve injury	Fewer

- **Maylard**: transects the **rectus muscles and sheath horizontally** to widen space; harder (muscle cutting, ligation of **inferior epigastric arteries**, which lie lateral to the rectus bellies).
- **Morbid obesity**: best incision unclear; Parkland prefers a **periumbilical midline vertical** incision.
- **Plastic barrier retractor (Alexis-O)** does **not** lower SSI.

Pfannenstiel steps

- Skin: low, slightly curved, at the **pubic hairline (~3 cm above the symphysis)**, **12–15 cm** long.
- **Superficial epigastric vessels** lie halfway between skin and fascia, several cm from midline → coagulate.
- Fascia has **two layers** (external oblique aponeurosis + fused internal oblique/transversus aponeuroses); incise individually. **Inferior epigastric vessels** lie lateral to the rectus, beneath the fused layer → at risk with far lateral extension.

- Separate fascia from rectus down to the symphysis and up (semicircle, radius **~8 cm**). **Meticulous hemostasis** ↓ hematoma and infection.
- Separate rectus and pyramidalis in the midline → transversalis fascia and peritoneum.
- **Enter peritoneum high** (lowers **cystotomy** risk); check the tented fold for omentum, bowel, bladder. Beware adhesions after prior surgery and a **high-riding bladder in obstructed labor**.

Midline vertical

- Begins **2–3 cm above the symphysis**; **12–15 cm** long.
- Open fascia in the **upper half of the linea alba** (avoids cystotomy), then extend up and down.

Joel-Cohen and Misgav Ladach

- Differ from Pfannenstiel-Kerr by **higher, straight incision** and **more blunt dissection**.
- **Joel-Cohen**: straight **10-cm** incision **3 cm below the anterior superior iliac spines**; subcutaneous tissue opened only 2–3 cm in the midline; fascia opened laterally under intact fat; rectus separated by fingers in opposition; peritoneum entered **sharply**.
- **Misgav Ladach**: same, but **peritoneum entered bluntly**.
- Benefits: **shorter operating time, less blood loss and pain**. Harder with rectus fibrosis or adhesions (repeat surgery).

6. Hysterotomy

Incision	Notes
Low transverse (Kerr, 1921)	Preferred. Easy repair, less bleeding, less bowel/omental adhesion; in the inactive segment → less likely to rupture later
Low vertical (Krönig, 1912)	Confined to the lower segment; occasional
Classical	Low vertical incision extended up into the active corpus
Fundal or posterior	For placenta accreta syndrome

Low transverse cesarean incision

- **Palpate for uterine rotation (dextrorotation)** and re-center the uterus, so the incision does not extend into the uterine artery.
- **Bladder flap**: vesicouterine peritoneum grasped and incised transversely (Figs 30-1, 30-2), then bladder separated bluntly/sharply (**usually ≤5 cm**).
 - Moves the bladder away and protects it if the incision extends down.
 - **Extend the dissection further if hysterectomy is anticipated.**
 - Dense adhesions → sharp dissection; **back-fill the bladder via Foley** to find its edge.
 - **Omitting the flap** shortens incision-to-delivery time, but data are limited.

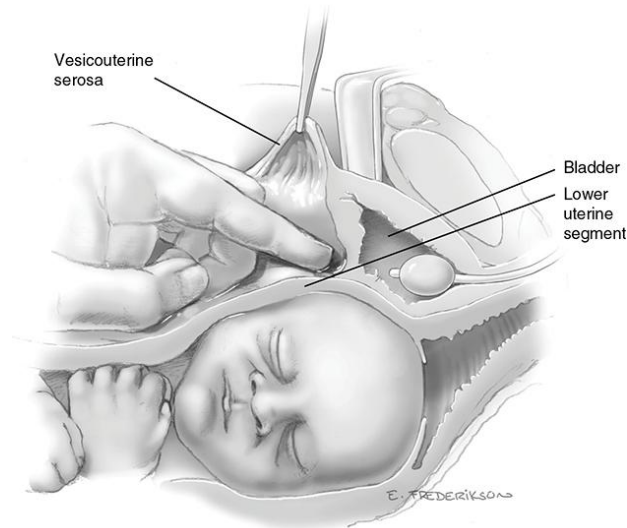


Figure 30-3. Bladder flap: blunt dissection of the bladder off the lower uterine segment (cross section).

Uterine incision

- Enter at the border between firm upper and flexible lower segment (often at the flap level). With **advanced or full dilation, place it higher** → otherwise extension into uterine arteries or incision into cervix/vagina.
- Scalpel **1–2 cm transverse** in the midline with **repetitive shallow strokes** (avoid fetal laceration), then finger entry.
- Extend by **blunt finger spreading** (lateral and slightly upward) or **cephalocaudal pull** (may ↓ extensions). Thick lower segment → **bandage scissors**, fingers shielding the fetus.
- **Blunt vs sharp expansion:** blunt → **fewer extensions, shorter surgery, less blood loss**; infection and transfusion rates the same.
- **Extension risk factors:** **occiput posterior, primary cesarean, advanced first or second stage.**
- **Placenta in the incision line** → detach or incise it, and **deliver quickly.**

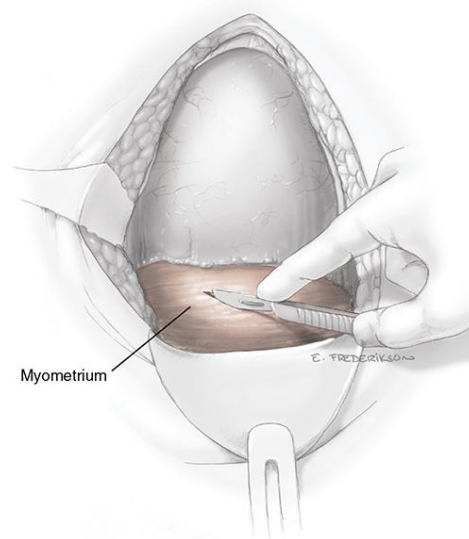


Figure 30-4. Myometrium incised with repeated shallow scalpel strokes to avoid cutting the fetal head.

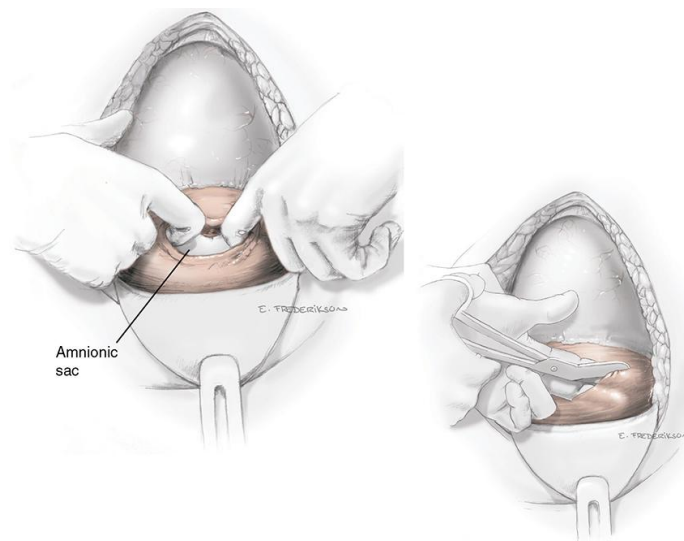


Figure 30-5. After entry, the incision is widened with fingers (or bandage scissors, inset).

Inadequate room → extended incisions

Extension	Shape	Consequence
J	One corner extended cephalad into the contractile myometrium	↑ blood loss; ↑ rupture risk with later TOLAC
U	Both corners extended	Same
T	Midline vertical extension	Same
Inferior extension	Down into the lower segment/cervix	Limited data: ↑ rupture; Parkland precludes TOLAC

- RCOG: insufficient evidence on TOLAC after a significant extension → individualize.
- A **classical or fundal incision greatly raises rupture risk** in later TOLAC → **tell the patient and document it** in the operative report.

Delivery of the fetus

- **Cephalic**: hand between symphysis and head, lift the head through the incision, **modest fundal pressure**.
- **Deeply impacted head** (long labor, CPD) → risks: extension, blood loss, **fetal skull fracture**.
 - **"Push"**: assistant pushes the head up vaginally (frog-leg position helps).
 - **"Pull"**: grasp the fetal legs and deliver as a **breech extraction** → **fewer uterine extensions**.
 - Pull by elevating both shoulders; or a **"fetal pillow"** (intravaginal balloon) — evidence limited.
- **Unlabored, unmolded, high head** in a thick lower segment → **forceps or vacuum**. Rotate to **occiput transverse** first.
- After the head: check for **nuchal cord**; rotate to OT (bisacromial diameter vertical); axial traction for anterior shoulder, then upward for posterior. **Avoid abrupt lateral force** (brachial plexus).
- **No routine suctioning**, even with meconium (AHA).
- **Delayed cord clamping** does not lower maternal Hb on postop day 1.
- **A person skilled in neonatal resuscitation must be present** (AAP/ACOG). Need for resuscitation after elective cesarean under neuraxial ≈ vaginal birth.
- **Skin-to-skin** in the OR; breastfeeding can start the day of surgery.

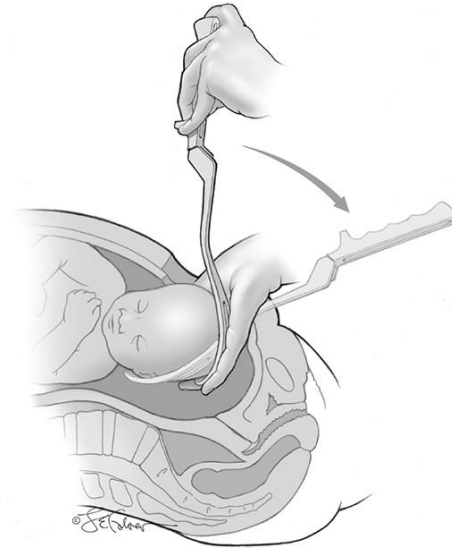


Figure 30-7. Forceps at cesarean: the first blade slides along the operator's palm to lie over the malar/parietal bone.

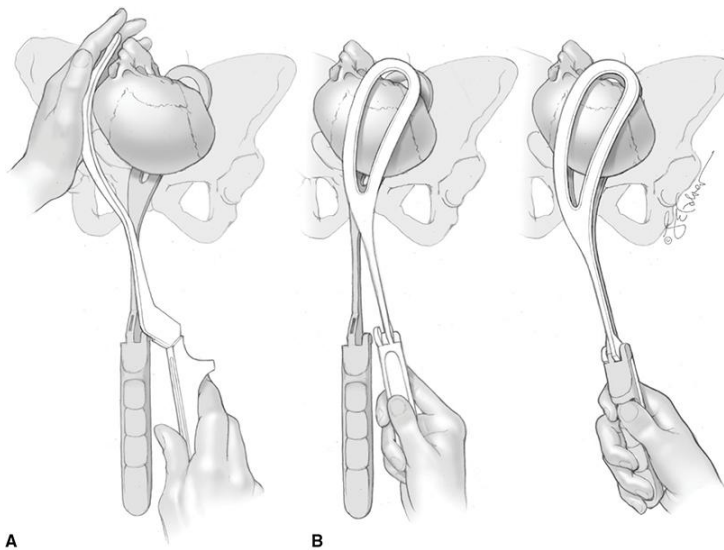


Figure 30-8. Second blade guided in via the posterolateral hysterotomy (A); handles locked, then traction up and out (B).

Delivery of the placenta

- Clamp briskly bleeding incision sites (**Pennington or ring forceps**).
- **Spontaneous delivery with gentle cord traction + fundal massage** → **less blood loss and infection** than manual removal.
- Inspect the placenta, wipe out the cavity. **Dilating the cervix** from above does **not** reduce infection or PPH.

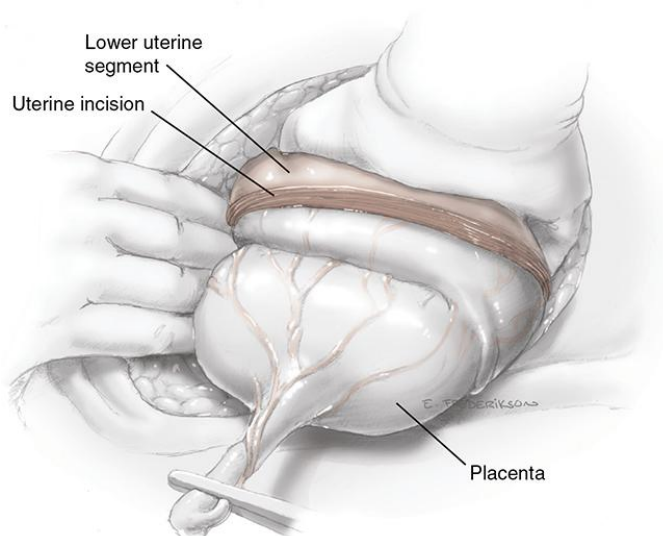


Figure 30-9. Placenta bulging through the incision; fundal massage aids spontaneous separation.

Uterotonic prophylaxis

- **Oxytocin 20 units (2 ampules) per liter crystalloid at 10 mL/min**, reduced once the uterus contracts. **Avoid undiluted boluses (hypotension).**
- **WHO: oxytocin is first-line.**
- Alternatives: **carbetocin** (long-acting, not in US), misoprostol, **methylergonovine, ergonovine**, Syntometrine (oxytocin + ergonovine).
- **Tranexamic acid** with oxytocin: ↓ EBL >1000 mL or transfusion (Sentilhes 2021), but **no reduction in secondary blood-loss clinical outcomes.**

Uterine repair

- **Exteriorize** (Parkland) → faster recognition of atony, easier view of bleeding/extensions and adnexa. **vs in situ: no difference in endometritis or transfusion (CORONIS)**; nausea, vomiting and pain similar.
- Insert a planned **IUD before closure**. Count before closing.
- **Closure:** one or two layers of **continuous 0 or No. 1 absorbable suture** (chromic catgut or polyglactin 910). **Suture type doesn't change later rupture risk.**
- **Single-layer:** faster; **no more infection or transfusion**; most studies: number of layers doesn't affect next-pregnancy complications. **Double-layer → slightly more scar niches at 3 months** (Stegwee 2021).
- **Parkland: one-layer, running locking chromic**, full thickness, from just beyond one end to just beyond the other. Persistent bleeding → extra layer or figure-of-8/mattress stitches.
- **Peritoneal (bladder flap) closure not needed** — no complications from omitting it.

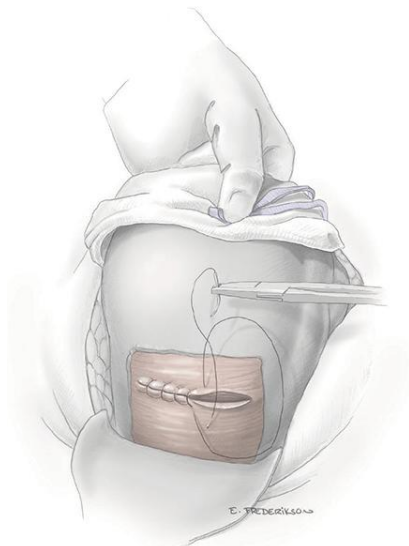


Figure 30-10. Hysterotomy closed with a running, locking suture.

Adhesions

- Form in the **vesicouterine space** and between uterus and anterior abdominal wall; **increase with each cesarean**.
- Lengthen incision-to-delivery and operating time; ↑ **cystotomy and bowel injury**.
- Reduce by gentle handling, hemostasis, less ischemia/infection/foreign body. **Peritoneal closure and adhesion barriers: no benefit**.

Abdominal closure

- Suction gutters and cul-de-sac. **Routine irrigation** in low-risk women → more nausea, **no less infection**.
- **Rectus fascia: continuous, nonlocking, delayed-absorbable** (monofilament if high infection risk).
- **Subcutaneous tissue: close if ≥ 2 cm thick** (↓ seroma/hematoma → ↓ wound infection/disruption). **Subcutaneous drain does not help**.
- **Skin:** subcuticular **4-0 delayed-absorbable** (poliglecaprone 25 or polyglactin 910), glue (2-octyl cyanoacrylate, equivalent for Pfannenstiel) or staples. Cosmesis and infection similar; **staples → more wound separation**; suturing takes longer.
- **Negative-pressure wound therapy** in obese women: evidence mostly **against** routine use.

Classical cesarean incision

Indications (avoid otherwise: the active upper segment is prone to rupture)

Maternal / access	Fetal
Densely adhered bladder	Transverse lie of a large fetus , especially with ruptured membranes and shoulder impacted
Leiomyoma in the lower segment	Back-down transverse lie (can't reach a leg through a low transverse incision)
Cervical cancer invading the cervix	Very small preterm breech (poorly developed lower segment; head entrapment)
Prior radical trachelectomy	Multiple fetuses (malpositioned or preterm)
Massive obesity blocking access	Large fetal malformation or conjoined twins
Anterior placenta previa , especially with accreta (incision may go higher or posterior; cephalic fetus delivered like a breech extraction)	

- **Technique:** start **as low as possible**, then extend cephalad with **bandage scissors** over the surgeon's fingers. Large myometrial vessels bleed profusely.

- **Closure (multilayer):** deep half with running **0 or No. 1 chromic**, then the outer half, then the **serosa** — avoids suture tearing through.

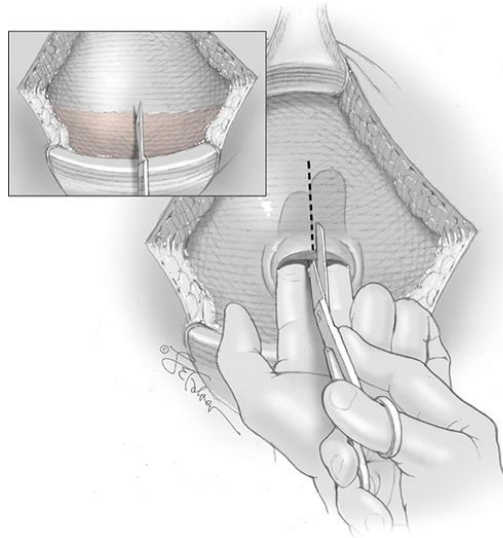


Figure 30-11. Classical incision: small low vertical cut, then scissors extend it cephalad over protecting fingers.

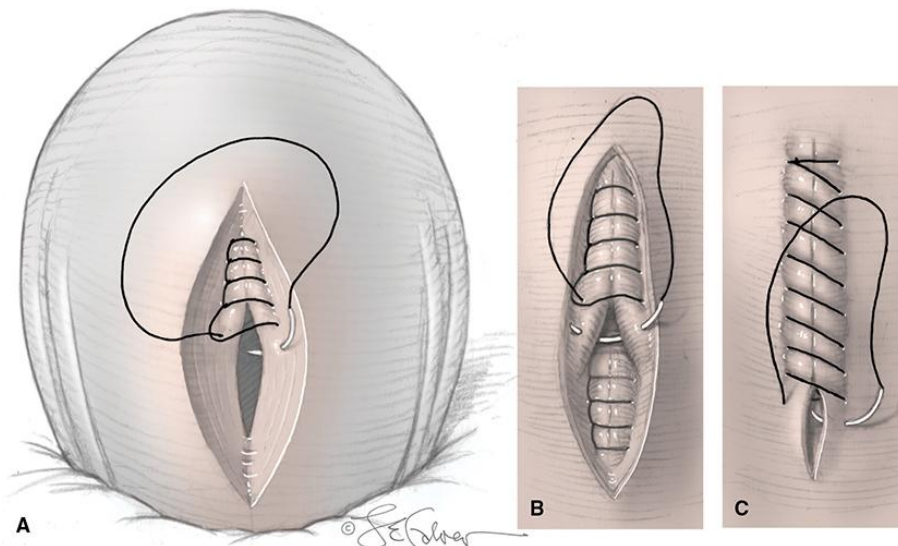


Figure 30-12. Classical closure in layers: deep half (A), superficial half (B), then serosa (C).

7. Peripartum Hysterectomy

Indications and outcomes

- **Most common reasons:** to stop or prevent hemorrhage from **intractable atony**, **surgical trauma/tears**, or **abnormal placentation**.
- US rate **~1 per 1000 births** and rising (Parkland **1.7 per 1000** over 25 years), driven by **rising cesarean rates** and their complications in later pregnancies.
- **½ to ⅔ total**, the rest supracervical.
- **Major complications:** blood loss and **urinary tract injury**.
- **Planned (elective) vs emergent:** planned → **less blood loss, fewer transfusions, fewer urinary tract complications**. Accreta preparation: Ch 43.

Technique

- Exposure with assistant's cephalad uterine traction + handheld retractors (a Balfour isn't needed initially).

- **Bladder flap as low as the cervix**; if hysterectomy is planned, **do the extended dissection before the hysterotomy**.
- **Placenta accreta with planned hysterectomy** → **leave the placenta in situ**. Bleeding hysterotomy → quick full-thickness closure or Pennington/sponge forceps.

Steps (each side, then contralateral):

1. **Round ligament** clamped, cut, ligated (0 or No. 1 suture) → open the anterior leaf of the broad ligament down to the bladder flap.
 2. **Perforate the posterior leaf** beneath the tube, utero-ovarian ligament and ovarian vessels.
 3. Clamp and cut **tube + utero-ovarian ligament**; lateral pedicle **doubly ligated** (medial clamp stays with the specimen).
 4. Divide the posterior leaf down toward the **uterosacral ligament**; dissect the bladder further (sharply if adherent).
 5. **Uterine vessels** clamped close to the uterus, divided, **doubly suture-ligated**. The **ureter passes beneath the uterine artery** → assistant pulls the uterus to the opposite side.
- **Profuse hemorrhage**: rapidly clamp and divide all pedicles first, ligate afterward.

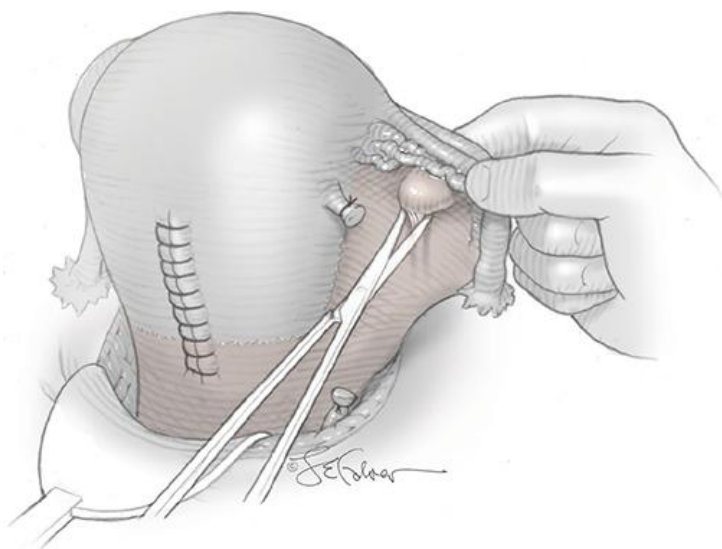


Figure 30-14. Posterior leaf of the broad ligament perforated beneath the tube, utero-ovarian ligament and ovarian vessels.

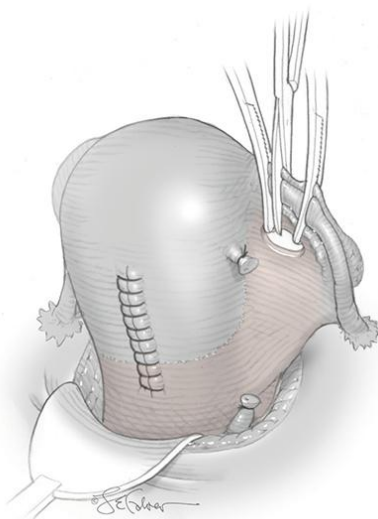


Figure 30-15. Tube and utero-ovarian ligament clamped and cut; lateral pedicle doubly ligated.

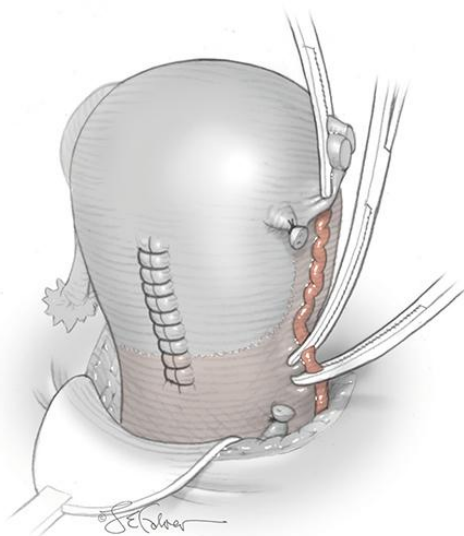


Figure 30-18. Uterine vessels clamped close to the uterus (ureter lies just beneath) and doubly ligated.

Total hysterectomy

- Often easier to **amputate the fundus first** and hold the cervical stump with Ochsner/Kocher clamps.
- Mobilize the bladder further → carries the **ureters caudad** and protects the bladder.
- **Cardinal and uterosacral ligaments** clamped with Heaney clamps **close to the cervix**, cut and ligated in steps down to the lateral vaginal fornix.
- **Finding a dilated, effaced cervix:** feel through the hysterotomy or a low anterior midline uterine incision (change the contaminated glove), or **place 4 clips/colored sutures at 12, 3, 6, 9 o'clock transvaginally** before a planned case.
- Clamp across the lateral vaginal fornix below the cervix, cut above it, confirm the whole cervix is out.
- **Vaginal cuff:** transfixing stitch at each angle + interrupted sutures for the middle; **attach the lateral fornices to the uterosacral ligaments** (prevents prolapse).
- Survey all pedicles for bleeding, avoiding the ureters.

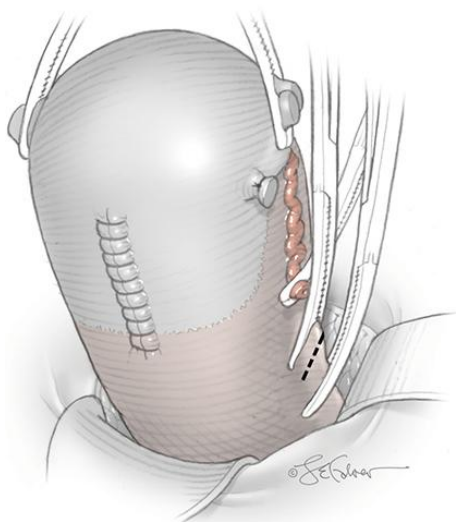


Figure 30-19. Cardinal ligaments clamped, cut and ligated close to the cervix.

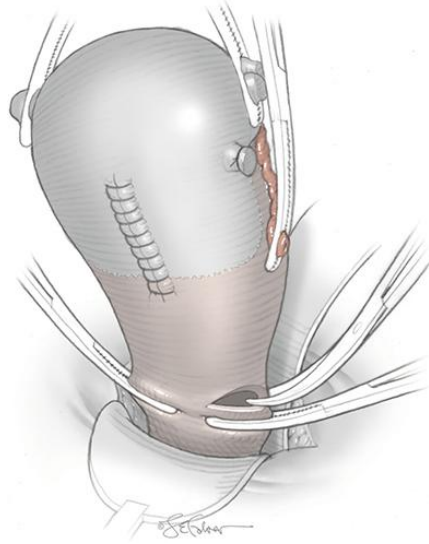


Figure 30-20. Curved clamp across the lateral vaginal fornix just below the cervix; tissue cut medially.

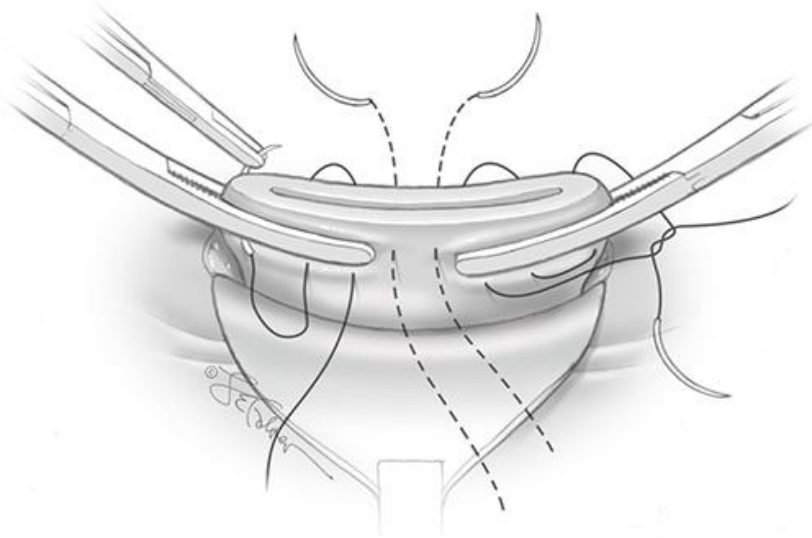


Figure 30-21. Vaginal cuff closed with lateral transfixing stitches plus interrupted midline sutures.

Supracervical (subtotal) hysterectomy

- Amputate the uterine body **just above the uterine artery ligation**; close the stump.
- **Often all that is needed to stop hemorrhage.** Preferred when a **shorter operation** is needed or **adhesions threaten the urinary tract.**

Salpingo-oophorectomy

- May be needed for hemostasis (large adnexal vessels close to the uterus). **Unilateral or bilateral oophorectomy in ~1/4 of cases** (Briery 2007) → include in counseling.

Urinary tract and bowel injury

Injury	Rate per cesarean
Bladder laceration	~2 per 1000
Ureteral injury	~0.3 per 1000
Bowel injury	~1 per 1000

Cystotomy

- Occurs during **bladder flap dissection, peritoneal entry** or **hysterotomy.**
- **Risk factors:** prior cesarean, adhesions, **emergency cesarean, cesarean hysterectomy** (especially accreta), **second-stage surgery.**

- Signs: **clear fluid gush**, visible **Foley bulb**. Confirm by retrograde filling (**dilute sterile infant formula** or **methylene blue saline**).
- **Dome most common**; the rest involve the **trigone**.
- **Before repair: confirm both ureteric jets** (through the cystotomy, or a separate extraperitoneal dome cystotomy if near the trigone).
- **Repair: 2–3 layers, running 3-0 absorbable/delayed-absorbable**; first layer inverts the mucosa, then muscularis; test with marker fluid.
- **Continuous bladder drainage 7–14 days** (↓ fistula). **No prophylactic antibiotics** needed. Cystogram may be skipped for a simple laceration **<1 cm**.
- Trigone/large injuries → specialist; catheterize ureters if jets unseen; stents may stay.
- **Unrecognized cystotomy**: hematuria, oliguria, pain, ileus, ascites, peritonitis, fever, **urinoma, fistula** → retrograde cystography or **CT cystography**; repair promptly.

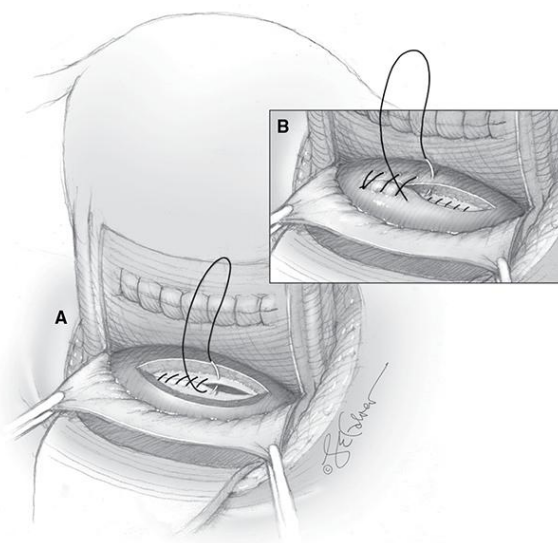


Figure 30-22. Cystotomy repair: mucosa inverted with 3-0 suture (A), then 1–2 muscularis layers (B).

Ureteral injury

- Risk: **cesarean hysterectomy (especially accreta)**, and repair of **extensions into the broad ligament or vagina**.
- Preop ureteral catheters: individualize (not Parkland practice). Lithotomy positioning allows cystoscopy.
- **Check with IV dye**:
 - **Methylene blue 50 mg IV over 5 min** — ⚠ risk of **methemoglobinemia with G6PD deficiency**.
 - **10% sodium fluorescein 25–50 mg IV** (bright yellow).
 - Look for brisk colored jets from each orifice (cystoscopy or cystotomy). **Sluggish jets may just be hypovolemia** → resuscitate.
- **Catheter test**: 4F–6F catheter up the orifice; most injuries are **at or below the pelvic brim (~13 cm from the orifice)**. Won't pass → kink, ligation or crush. **Catheter seen in the abdomen** → transection.
- **Management**:
 - Kinked/ligated → release the suture. Crush → **stent** to prevent stricture.
 - **Stents 4F–7F, 20–30 cm** (24–26 cm for most adults), **double-J/double-pigtail**. **Foley 7–10 days, stents out by cystoscopy after 14 days**.
 - Devascularization, thermal injury, transection → **ureteroneocystostomy** if tension-free; more proximal → **ureteroureterostomy, psoas hitch, Boari flap**.
- **Unrecognized**: like cystotomy + **CVA tenderness; CT urography first**.

Bowel injury

- **Serosal tears** are weak points → may perforate if obstruction develops. Oversew a few with fine suture; larger injuries → general surgeon or gynecologic oncologist.

8. Postoperative Care

Euvolemia

- ERAS: **euvolemic replacement** with lactated Ringer or similar crystalloid $\pm$ 5% dextrose.
- **Blood loss: ~1000 mL** for uncomplicated cesarean; **~1500 mL** for elective cesarean hysterectomy (more if emergent).
- A woman with **Hct $\geq$ 30%** and normal pregnancy volume expansion tolerates **up to 2000 mL** loss.
- Blood loss is often **underestimated** (vaginal loss, concealed uterine blood) → check Hct intra/postop as needed.

Recovery suite

- Watch **vaginal bleeding $\geq$ 1 hour** and palpate the fundus (firmly contracted).
- Transfer when: **minimal bleeding, stable vital signs, adequate urine output.**

Hospital care until discharge

Analgesia

Option	Detail
Neuraxial (intrathecal/epidural) morphine	12–24 h relief; preferred over intermittent parenteral opioid. Dose-related sedation and respiratory depression (monitoring protocol); pruritus, nausea
NSAIDs alternating with acetaminophen	Added relief
Oxycodone 2.5–5 mg q4h	Breakthrough; max 30 mg/d (neonatal sedation)
IM meperidine 50–75 mg or IM morphine 10–15 mg q3–4h	Severe breakthrough pain
TAP block	If no neuraxial anesthesia
PCA morphine	1 mg, 6-min lockout, max 30 mg/4 h ; 2-mg booster $\times$ 2 allowed

Monitoring and fluids

- Assess **hourly for 4 h**, then every 4 h (vitals, uterine tone, urine, bleeding). Encourage deep breathing and coughing.
- **Hct the morning after surgery**; sooner if big loss or hypotension, tachycardia, oliguria. Significant drop → repeat and look for the cause. **Iron preferred over transfusion** if bleeding has stopped.
- Maintenance IV fluid until eating (she mobilizes her expanded extravascular fluid).
- **Urine $<$ 30 mL/h → reassess promptly** (unrecognized bleeding, **oxytocin antidiuretic effect**).
- Unscheduled cesareans may have **contracted extracellular volume** (severe preeclampsia, sepsis, vomiting, long labor, bleeding) → keep in recovery longer.

Bladder and bowel

- **Foley removal: SOAP 6–12 h** (retention risk with long-acting neuraxial); ERAS **immediately**.
- **Urinary retention ~5%**; risks: **failure to progress in labor**, postop narcotics.
- **Early feeding** within hours in uncomplicated cases.
- **Ileus** (distention, colic, no flatus/stool): some degree after every laparotomy but negligible after most cesareans.
 - Plain radiograph diagnoses only **50–60% of SBO** → use as triage. The postpartum uterus can compress the rectosigmoid and **mimic distal colonic obstruction**.
 - **CT with IV + oral contrast** is more accurate (also for unrecognized bowel injury).
 - Treat: IV fluids, **correct electrolytes**; **NG tube only for persistent vomiting or severe distention**.
 - Prevention: minimal bowel handling, avoid fluid excess or hypovolemia, short surgery; **chewing gum** ($\geq$ 3 sessions/day, 15–60 min) speeds bowel recovery.

Ambulation and wound

- **Early ambulation** lowers VTE risk. Abdominal binder optional (evidence conflicting).
- Remove dressing at **24 h** (6 h is also fine); inspect daily. **Showering by day 3.**
- **Staples out on day 4** (tape strips for 1 week); keep **7–10 days** if superficial separation is a concern.

Hospital discharge

- Average stay **3–4 days**. Earlier discharge is feasible for selected women, but **discharge <96 h ↑ neonatal readmission** (plan jaundice checks).
- First week: self-care and newborn care with help. **Drive** when pain allows emergency braking and no narcotics.
- **Intercourse resumed: 44% by 6 wk, 81% by 3 months, 97% by 1 year.** Sexual function after the puerperium = vaginal birth.
- Return to work: **6 weeks** commonly cited; FMLA allows up to **12 weeks**.

Quick-Recall: Cesarean Decisions at a Glance

Question	Answer
Earliest elective / CDMR cesarean	39 weeks
Antibiotic and timing	Cefazolin 1 g (2 g ≥80 kg; 3 g ≥120 kg) within 60 min before incision
Add for labor/ROM	Azithromycin 500 mg IV
β-lactam anaphylaxis	Clindamycin 900 mg + aminoglycoside 5 mg/kg
Redose antibiotic	EBL >1500 mL or >3 h
VTE prophylaxis (ACOG)	Pneumatic compression before surgery
Preferred skin prep	Chlorhexidine-alcohol
Preferred incision	Pfannenstiel + low transverse hysterotomy
Fastest entry	Midline vertical
Uterine closure layers	One or two — outcomes similar; double → slightly more niches
Close peritoneum?	No (no benefit)
Close subcutaneous fat?	Yes if ≥2 cm
Staples vs suture	Staples → more wound separation
Impacted head	Push (vaginal) or pull (breech extraction) — pull ↓ extensions
High unlabored head	Forceps or vacuum
Placenta	Spontaneous + cord traction + massage (less blood loss/infection)
Uterotonic	Oxytocin 20 U/L at 10 mL/min ; no bolus
Hysterectomy for accreta	Leave placenta in situ ; extended bladder dissection before hysterotomy
Cystotomy	2–3 layers 3-0 , Foley 7–14 d
Ureter check	Methylene blue 50 mg IV (not in G6PD) or fluorescein
Unrecognized ureter injury	CT urography

Key Numbers to Memorize

- Cesarean rate **31.9%** (2018); **>85%** for 4 indications; VBAC **28% → 13.3%**
- Maternal mortality **2.2 vs 0.2 per 100,000** (cesarean vs vaginal); morbidity **2×**
- Fetal injury **1%** (elective **0.5%**); CDMR **~5%**
- Maturity criteria: US **<20 wk**, FHT **30 wk**, positive test **36 wk** ago

- NPO solids **6 h**, clears **2 h**; Bicitra **30 mL**
- VTE after cesarean **8 per 10,000**
- Pfannenstiel **3 cm above symphysis, 12–15 cm**; Joel-Cohen **10 cm, 3 cm below ASIS**
- Bladder flap **≤5 cm**
- Blood loss: cesarean **~1000 mL**; cesarean hysterectomy **~1500 mL**; tolerates **2000 mL** if Hct ≥30%
- Peripartum hysterectomy **~1 per 1000** births; oophorectomy in **1/4**
- Bladder injury **2/1000**, ureter **0.3/1000**, bowel **1/1000**; ureter injury **~13 cm** from orifice
- Urine output alarm **<30 mL/h**; urinary retention **~5%**
- Oxycodone max **30 mg/d**; PCA morphine **1 mg / 6 min / 30 mg per 4 h**
- Stay **3–4 days**; staples out **day 4**; intercourse **44% / 81% / 97%** at 6 wk / 3 mo / 1 y